

OPERATING SYSTEMS

Fundamentals, Functions & Architecture

Computer System Components

Hardware

- CPU
- Memory
- I/O Devices

Software

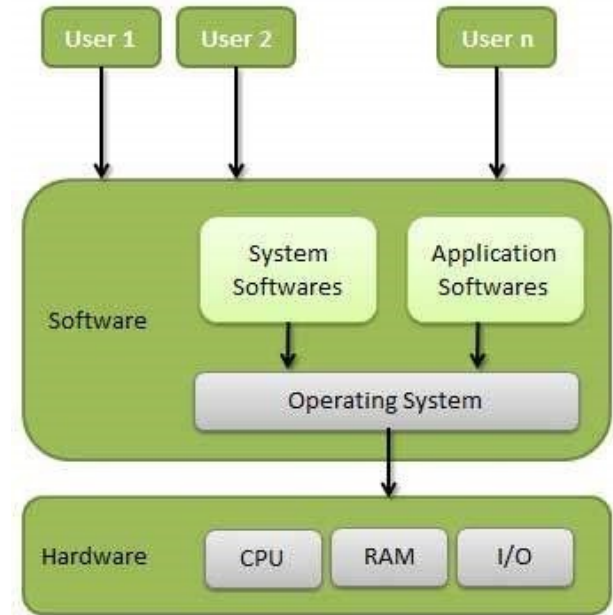
- Compilers
- Text Editors
- Database Systems

Operating System (Kernel)

Interface between users/applications and hardware

What is an Operating System?

- A program running at all times on the computer (kernel)
- Acts as interface between users and computer hardware
- Manages hardware and controls resource usage
- Provides efficient and convenient system usage



Operating System Objectives

Convenience

Make the computer system easier to use

Efficiency

Use system resources in an efficient manner

Evolvability

Support testing, development & addition of new components

OS as an Extended Machine

Hides Complex Hardware Details

Example: Disk Access

Without OS (Complex):

- Specify block address
- Calculate sector numbers
- Handle inter-sector gaps

With OS (Simple):

Just specify:

- File name
- Data location

OS as a Resource Manager

OS Responsibilities:

- Keep track of allocated resources
- Grant resource requests when available
- Account for resource usage
- Resolve conflicts between different programs/users

Resource Multiplexing

Time Multiplexing

Resources are shared turn-by-turn

- Resource allocated to User A for T1
- Then to User B for T2
- Then to User C for T3
- Example: CPU scheduling

Space Multiplexing

Resources split into portions

- Each user gets a portion simultaneously
- Example: Main memory
- Example: Disk storage
- OS allocates and deallocates space

OS: User Point of View

Stand-Alone

Ease of use

Personal computers

Networked

Resource utilization

Multiple users competing

Handheld

Individual usability

Mobile devices

Embedded

Reliable operation

No user intervention

OS: System Point of View

The OS has two primary roles:

Resource Allocator

- Manages CPU time
- Allocates memory
- Controls I/O devices
- Ensures efficient & fair distribution

Control Program

- Manages user programs
- Prevents errors
- Prevents improper system use
- Controls I/O operations

Key Takeaways

- OS is the bridge between users and hardware
- Achieves convenience, efficiency, and evolvability
- Functions as an extended machine and resource manager
- Uses time & space multiplexing for resource sharing
- Adapts to different computing environments